

SONiX 32-Bit Cortex-M0 Micro-Controller

1M ADC Example Code

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AMENDENT HISTORY

Version	Date	Description
1.0	2024/11/26	1. First version.

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1 OVERVIEW

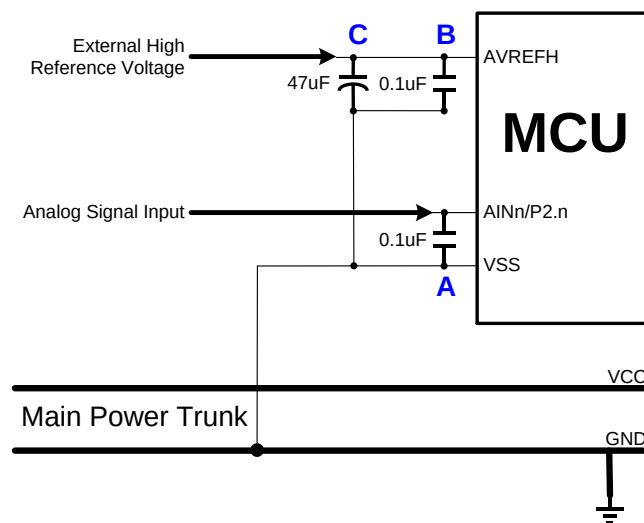
This document provides our customers the C program examples of ADC.

2 FEATURE

- ADC Type: Analog to Digital Convertor function.
- Sampling rate up to 1Msps
- Support auto-conversion mode
- Support ADB watchdog
- Use ARM-compiler V6.12

3 HW

- The starter kit board of vendor desired MCU.
- The ADC input signal must be through a 0.1uF capacitor. The 0.1uF capacitor is set between ADC input pin and VSS pin, and must be on the side of the ADC input pin as possible. Don't connect the capacitor's ground pin to ground plain directly, and must be through VSS pin. The capacitor can reduce the power noise effective coupled with the analog signal.



4 EXAMPLE CODE

4.1 ADC Initial

```

287 /* *****
288  * Function      : ADC_FuncInit
289  * Description  : Initialization of ADC
290  * Input       : bPCLKDiv - ADC_DIV1, ADC_DIV2, ..., ADC_DIV32
291  *             bADCLen - ADC_8BIT, ADC_12BIT
292  *             bCHMode - Single_Channel, Multiple_Channel
293  *             bSCMode - Single_Mode, Continuous_Mode
294  * Output      : None
295  * Return      : None
296  * Note       : None
297  * *****
298 void ADC_FuncInit(uint8_t bPCLKDiv, uint8_t bADCLen, uint8_t bCHMode, uint8_t bSCMode)
299 {
300     __ADC_ENABLE_HCLK;                                //Enables HCLK for ADC
301
302     SN_ADC->ADM_b.AVREFHSEL = ADC_AVREFHSEL_INTERNAL;    //Set ADC high reference voltage source from internal reference
303     SN_ADC->ADM_b.VHS = ADC_VHS_INTERNAL_2V;           //Set ADC high reference voltage source as Internal 2V
304
305     SN_ADC->ADM_b.GCHS = ADC_GCHS_EN;                  //Enable ADC global channel
306
307     SN_ADC->ADM_b.ADLEN = bADCLen;                    //Set ADC resolution = 12-bit
308
309     SN_ADC->ADM_b.ADCKS = bPCLKDiv;                   //ADC_CLK = ADC_PCLK/32
310
311     SN_ADC->CONVCTRL_b.SCMODE = bSCMode;              //Set mode
312
313     SN_ADC->CONVCTRL_b.CH = bCHMode;                 //Set converting channel
314
315     SN_ADC->ADM_b.ADENB = ADC_ADENB_EN;              //Enable ADC
316
317     UT_DelayNx10us(10);                                //Delay 100us
318
319     SN_ADC->ADM1_b.ACS = ENABLE;                      //Calibration start
320     while (SN_ADC->ADM1_b.ACS == 1);
321
322     SN_ADC->ADM1_b.CALIVALENB = ENABLE;              //ADC conversion with calibration value
323 }

```

Call the function to set the ADC reference voltage source, ADC resolution, and ADC conversion time.

Example: Set ADC high reference voltage source as Internal 2V
 Set ADC resolution as 12 bit
 ADC conversion time = PCLK / 32

```

345  /*****
346  * Function      : ADC_InterruptInit
347  * Description   : ADC Interrupt control of ADC channels
348  * Input        : bEOCALen - Enable bit refer to end of calibration
349  *               bOVRen - Enable bit refer to overrun
350  *               bAWWen - Enable bit refer to window watchdog
351  *               bEOSen - Enable bit refer to end of sequence
352  *               bADCCh - Enable bit refer to ADC channels
353  * Output       : None
354  * Return        : None
355  * Note         : None
356  *****/
357  void ADC_InterruptInit(uint8_t bEOCALen, uint8_t bOVRen, uint8_t bAWWen, uint8_t bEOSen, uint16_t bADCCh)
358  {
359      SN_ADC->IE = ((bEOCALen << 27) | (bOVRen << 26) | (bAWWen << 25) | (bEOSen << 24) | (bADCCh));
360      ADC_NvicEnable(); //Enable ADC NVIC interrupt
361  }

```

Call the function to control over which A/D channels generate an interrupt when a conversion is complete.

4.2 ADC Example

```

72  ADC_SingleCH_SingleMode_Example();
73  ADC_SingleCH_ContinuousMode_Example();
74  ADC_MultipleCH_SingleMode_Example();
75  ADC_MultipleCH_ContinuousMode_Example();
76  ADC_Window_Watchdog_Example();

```

There are 5 types of examples: single-channel single mode, single-channel continuous mode, multi-channel single mode, multi-channel continuous mode and ADB watchdog example.

5 RESULT

5.1 ADC Type

The ADC converter result can be observed from the Watch window.

The screenshot shows a C code editor with the following code:

```

81  /******
82  * Function   : ADC_SingleCH_SingleMode_Example
83  * Description : Demo code of ADC Single channel + Single mode
84  * Input      : None
85  * Output     : None
86  * Return     : None
87  * Note       : None
88  *****/
89  void ADC_SingleCH_SingleMode_Example(void)
90  {
91      volatile uint16_t wADC_ChSelect = ADC_CHS_AIN3;
92
93      __ADC_INTERRUPT_FLAG_CLEAR;
94
95      ADC_FuncInit(ADC_DIV32, ADC_12BIT, Single_Channel, Single_Mode);
96      ADC_InterruptInit(EOCALIE_DIS, OVRIE_DIS, AWVIE_DIS, EOSIE_DIS, wADC_ChSelect);
97
98      while (1)
99      {
100         if (ADC_Convert(wADC_ChSelect, ADC_FUNCTION_MODE1) == TRUE)
101         {
102             hwADC_Value[0] = ADC_Read();
103         }
104         else
105         {
106             while (1); //Fail
107         }
108         __ADC_INTERRUPT_FLAG_CLEAR;
109     }
110 }

```

The Watch window (Watch 1) displays the following data:

Name	Value	Type
hwADC_Value	0x20000004	hwADC_Value
[0]	0x0C08	ushort
[1]	0x0000	ushort
[2]	0x0000	ushort
[3]	0x0000	ushort
[4]	0x0000	ushort
[5]	0x0000	ushort
[6]	0x0000	ushort
[7]	0x0000	ushort
[8]	0x0000	ushort
[9]	0x0000	ushort
[10]	0x0000	ushort
[11]	0x0000	ushort
[12]	0x0000	ushort
[13]	0x0000	ushort
[14]	0x0000	ushort
[15]	0x0000	ushort
[16]	0x0000	ushort
[17]	0x0000	ushort
[18]	0x0000	ushort
[19]	0x0000	ushort
[20]	0x0000	ushort
[21]	0x0000	ushort

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